

Lists

SPECIAL NOTE

- This presentation consists of some theory slides too!
  - This is for learning purposes
- Additional reading
  - https://www.tutorialspoint.com/python/python_lists.htm

What is a List?

- It’s a compound data type
- No need to have same data type
- Possible to apply general operations on top of lists
  - Indexing
  - Adding
  - Slicing, etc.

Graphical Representation

Data Items

MyList

"IIT"	2017	"Python"	75.00	"Adios"
0	1	2	3	4

Data Location

MyList [0] ?? IIT
MyList [3] ?? 75.00

Some List Operations

- Display items
- Update items
- Delete items
- Reverse items
- Sort items

Sample Program & Output

```
# Create Variables
MyList = ["IIT", 2017, 'Python', 75.0, "Adios"]

# Diplay values
print (MyList)
print (MyList [0])
print (MyList [3])

# Modify values
MyList [0] = "Foundation"
print (MyList)
print (MyList [0])

MyList [3] = "Hello"
print (MyList)
print (MyList [3])

# Deleting items
del MyList[1]
print (MyList)
```

```
['IIT', 2017, 'Python', 75.0, 'Adios']
IIT
75.0
['Foundation', 2017, 'Python', 75.0, 'Adios']
Foundation
['Foundation', 2017, 'Python', 'Hello', 'Adios']
Hello
['Foundation', 'Python', 'Hello', 'Adios']
```

Sample Program & Output

```
# Create Variables
MyList = [100,200,550,200,150]
List2 = ["hi","how","are","you"]

# Some operations
print (MyList)
print (MyList[1:4])
print (len(MyList))

NewList = MyList + List2 # Join lists
print (NewList)

NewList = MyList * 2 # Multiply lists
print (NewList)
```

```
print (111 in MyList) # Is 111 inside MyList

for Value in MyList: # Display all elements
    print (Value)

print (MyList.count(200)) # How many times 200 occurs

print (MyList)
MyList.reverse() # Reversing elements
print (MyList)

MyList.sort() # Sort ascending
print (MyList)

MyList.sort(reverse=True) # Sort descending
print (MyList)
```

Output

```
[100, 200, 550, 200, 150]
[200, 550, 200]
5
[100, 200, 550, 200, 150, 'hi', 'how', 'are', 'you']
[100, 200, 550, 200, 150, 100, 200, 550, 200, 150]
False
100
200
550
200
150
2
[100, 200, 550, 200, 150]
[150, 200, 550, 200, 100]
[100, 150, 200, 200, 550]
[550, 200, 200, 150, 100]
```